INT 453: Digital Multimedia

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Lecture 4

Digital Multimedia Databases



Agenda

- What is MM data?
- MMDB & MMDBMS
- Multimedia databases characteristics.
- MMDB models
- MMDB Application areas &Multimedia federated database system
- Multimedia database Requirements
- Issues and challenges.
- Audio-Visual Data and Object Linking and Embedding (OLE)
- SQL3: Oracle Large Objects

Multimedia Database

Multimedia Data

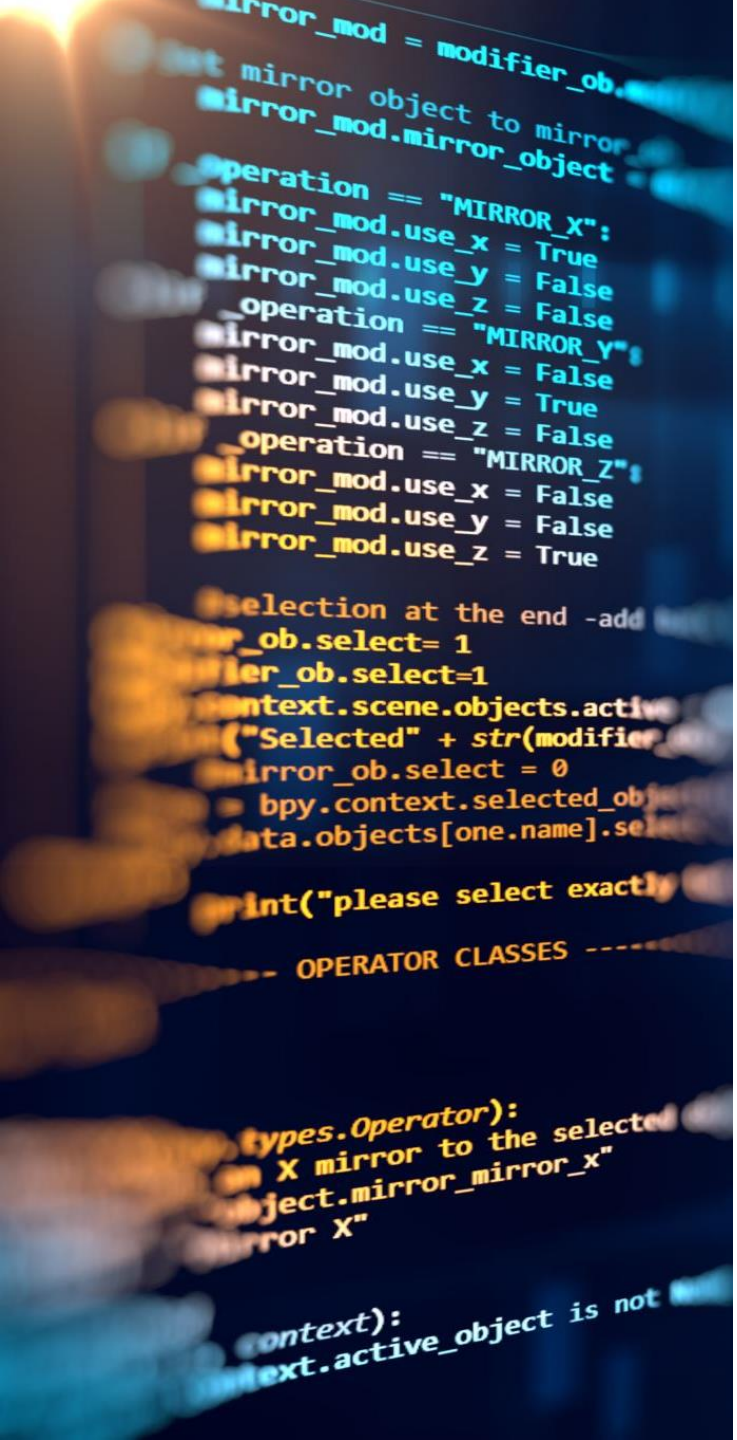
The multimedia data include one or more primary media data types such as:

- text,
- images,
- Graphic objects (including drawings, sketches and illustrations)
- animation sequence,
- audio and video.



MMDB & MMDBMS

- A **Multimedia database** (**MMDB**) is a collection of related multimedia data.
- A **Multimedia Database Management System** (**MMDBMS**) is a framework that manages different types of data potentially represented in a wide diversity of formats on a wide array of media sources. It provides support for multimedia data types, and facilitates for creation, storage, access, query , and control of a multimedia database.



Multimedia database characteristics

A true MMDBMS should be able to:

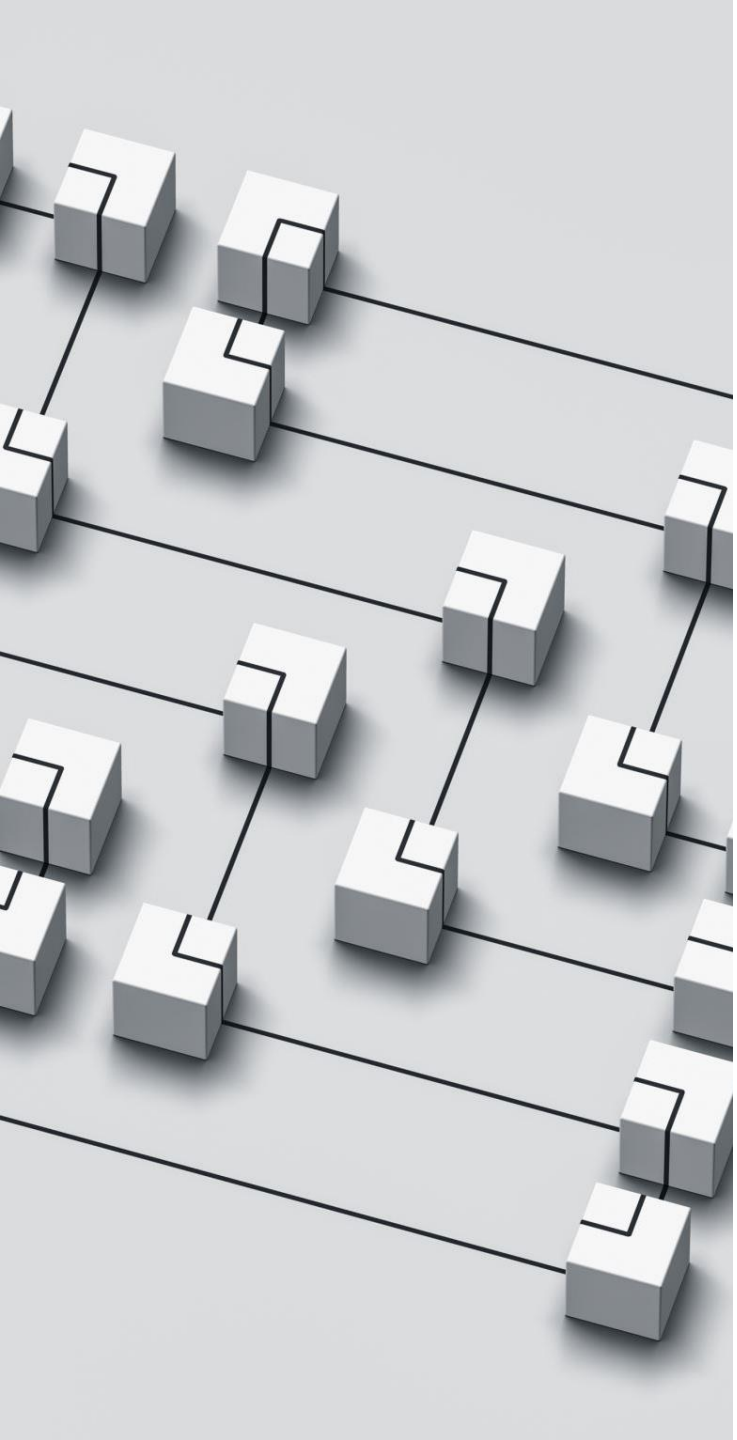
1. Operate (create, update, delete, retrieve) with at least all the audio-visual multimedia data types (text, images, video, audio) and perhaps some non-audio-visual types (e.g. olfactory, taste, haptic)
2. Fulfill all the standard requirements of a DBMS i.e. data integrity, persistence, transaction management, concurrency control, system recovery, queries, versioning, data integrity, data security, etc.
3. Manipulate huge data volumes (virtually no restriction concerning the number of multimedia structures and their size).



Multimedia database characteristics

A true MMDBMS should be able to:

4. Retrieve multimedia data based on their content (attributes, features and concepts)
8. MMDB May include advanced features such as character recognition, voice recognition, speech recognition (e.g. IBM ViaVoice, VoiceXML) and image analysis.
9. Many hypermedia systems are based on relational or object-relational database technologies e.g. Active Server Pages, ColdFusion, Java Server Pages.



Multimedia Database Models

1. Object-Oriented Databases
2. Object-relational databases
3. Document management systems / content management systems



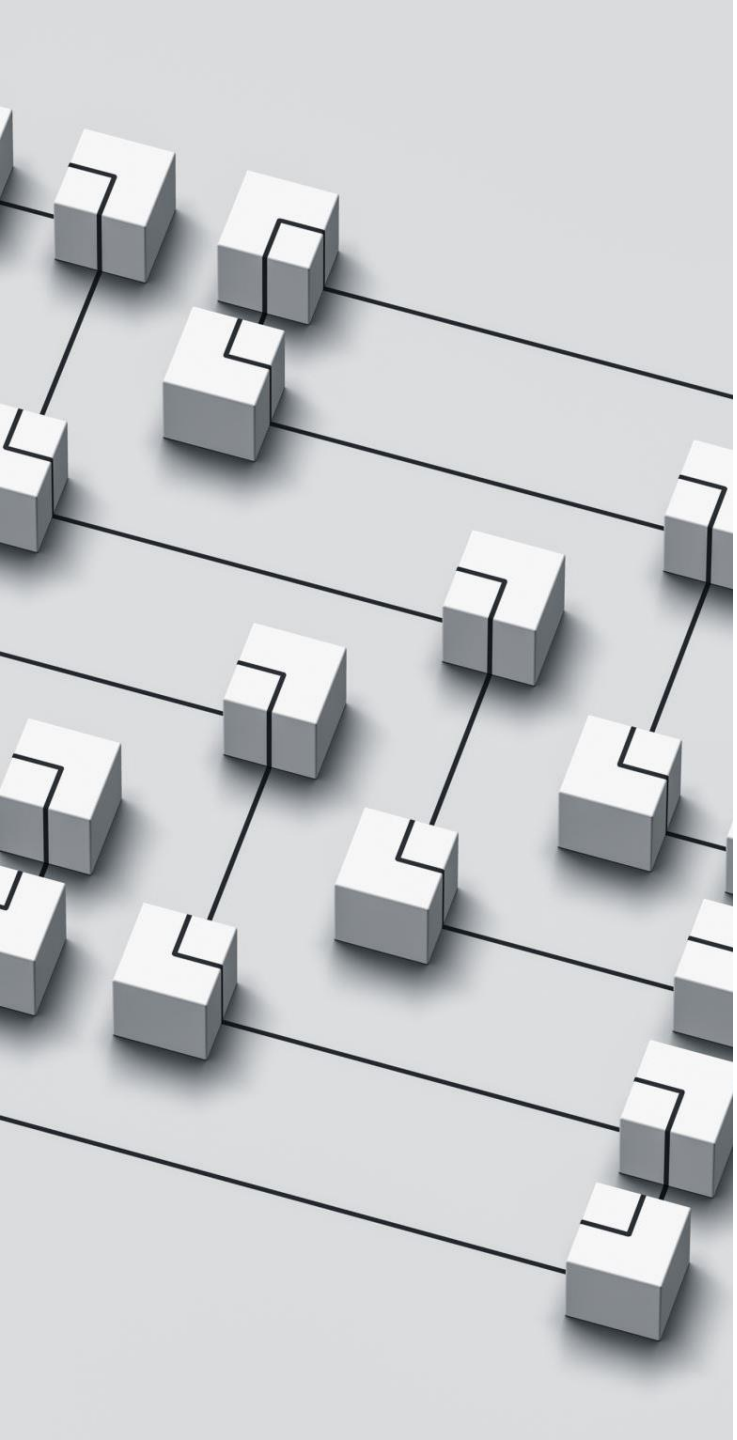
Multimedia Database Models

Object-Oriented Databases

- Handle all kinds of multimedia data in specialized classes
- Quite low market share in database technology

Object-relational databases

- Hybrid combination of features of RDBMS and OODB
- Fully sql-compliant, with added support for complex data types via object definition language (ODL)



Multimedia Database Models

Document management systems / content management systems

- Specialized systems for managing documents/content of many different types.
- Automate workflow, pages/reports generated automatically.
- Very popular for web sites where content changes rapidly e.G. Kenny's bookshop (irish times (ireland.Com).

MMDB Application areas

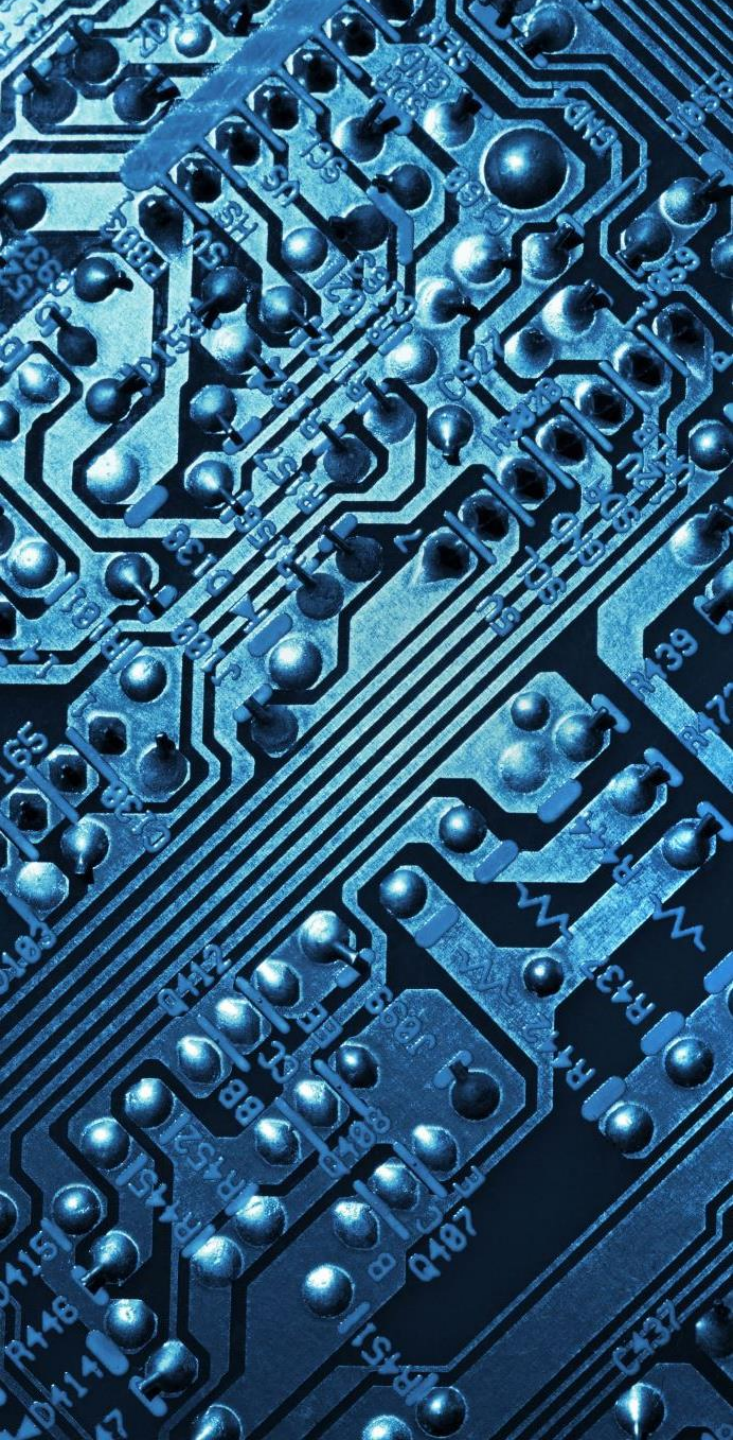
- Digital Libraries
- News-
- Video-
- Music
- G
- Tele

Multimedia federated database system
Is the process of integration of digital smell, taste and touch in a MMDBMS will most likely take the form of a– i.e. multiple specialised databases in an interconnected system architecture.



Multimedia database Requirements

- *Integration*
  - Data items do not need to be duplicated for different programs
- *Data independence*
  - Separate the database and the management from the application programs
- *Concurrency control*
  - allows concurrent transactions



Multimedia database Requirements

- *Persistence*
  - Data objects can be saved and re-used by different transactions and program invocations
- *Privacy*
  - Access and authorization control
- *Recovery*
  - Failures of transactions should not affect the persistent data storage
- *Query support*
  - Allows easy querying of multimedia data

What are Issues and challenges

- Multimedia data often require a large storage

Since, multimedia data consists of a variety of media formats or file representations including TIFF, BMP, PPT, IVUE, FPX, JPEG, MPEG, AVI, MID, WAV, DOC, GIF, EPS, PNG, etc. the data size of multimedia is large such as video; therefore,

- Multimedia database consume a lot of processing time, as well as bandwidth.
- Some multimedia data types have implications in storage and manipulation

MM including video, audio, and animation sequences have have implications on their storage, manipulation and presentation, but images, video and graphics data have spatial constraints in terms of their content.

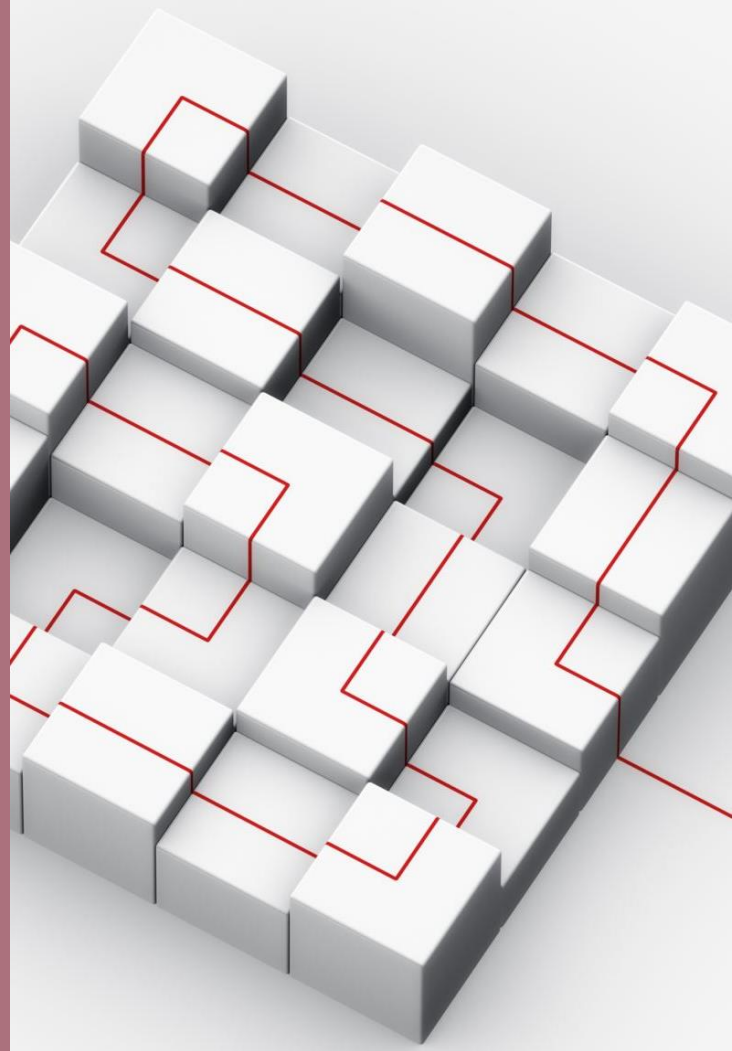


Audio-Visual Data

1. Discrete data: data not depending on a time scale are called discrete or static data
2. Continuous data: data depending on a time scale are called continuous or dynamic data
3. **Text, graphics and images are discrete data, while audio and video are continuous**
4. Continuous data are more complex than discrete data - demanding the use of much better compression/decompression algorithms and more sophisticated operations for their interpretation and manipulation.

Object Linking and Embedding (OLE)

- Microsoft's first architecture for integrating files of different types.
- Each file type in Windows is associated with an application
- It is possible to place a file of one type inside another, either by
  - wholly embedding a separate copy of the file,
  - or by placing a link to the actual file





SQL3: Oracle Large Objects

SQL3 support three large object types:

- BLOB - The BLOB domain type stores unstructured binary data in the database. BLOBs can store up to 4GB of binary data.
- CLOB – The CLOB domain type stores up to 4GB of single-byte character set data
- NCLOB - The NCLOB domain type stores up to 4GB of fixed-width and varying width multi-byte national character set data

Review Questions

Describe what is meant by:

- MM data
- **MMDB & MMDBMS**
- Multimedia databases characteristics.
- MMDB models
- MMDB Application areas &Multimedia federated database system
- **Multimedia database** Requirements
- Issues and challenges.
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Questions

Thank you